



ICT AND ITS CURRENT STATE

WHAT IS EMPOWERMENT?

Empowerment means:

- Giving someone the ability, confidence, or power to do something
- Helping people become stronger and more independent

Example:

Teaching someone how to use a computer empowers them to apply for a job online.

WHAT ARE TECHNOLOGIES?

Technologies are:

- Tools, systems, and devices that help us solve problems and do tasks easier
- Includes gadgets, apps, websites, and digital platforms

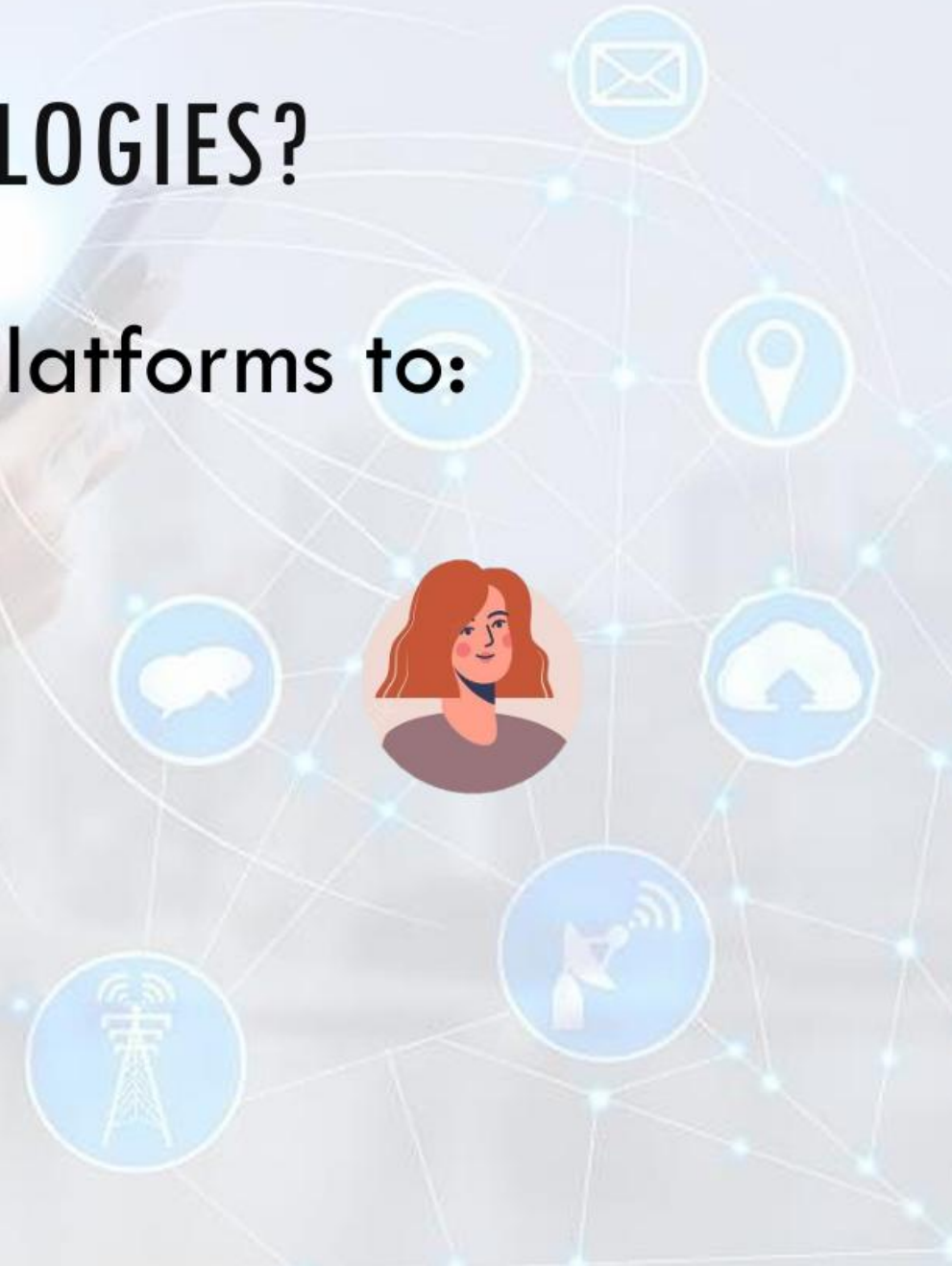
Examples:

Smartphones, internet, Zoom, Google Docs, Canva, social media

WHAT IS EMPOWERMENT TECHNOLOGIES?

It is the use of digital tools and platforms to:

- Communicate effectively
- Create content
- Learn new skills
- Share ideas
- Solve real-world problems



Empowerment Technologies helps students:

- ✓ Use tech responsibly and effectively
- ✓ Create digital outputs (e.g., posters, presentations, websites)
- ✓ Protect themselves online
- ✓ Understand how tech can improve lives





**Empowerment Technologies =
Learning to use technology to
make a difference.**





How information and
communication technology
affect our daily lives?



What was your ICT experience
so far?



INFORMATION COMMUNICATION TECHNOLOGY (ICT)

an umbrella term referring to communication hardware or software that allows the access, storage, transmission, and manipulation of information

INFORMATION COMMUNICATION TECHNOLOGY (ICT)

an umbrella term referring to communication hardware ← allows the access, storage, and manipulation of information

radio, television, communication devices (smart and non-smart phones), desktop computers, laptops, and network hardware and technologies

INFORMATION COMMUNICATION TECHNOLOGY (ICT)

an umbrella
communi
allows th
and mar

file transfer
protocol, messaging
software and e-mail

application or
program designed
to pass information
from one system to
another.

to
→ software that
transmission,
nation

INFORMATION COMMUNICATION TECHNOLOGY (ICT)

“All kinds of electronic systems used for broadcasting telecommunications and mediated communications”

INFORMATION COMMUNICATION TECHNOLOGY (ICT)

Computer Technology

storing and processing
information in digital
form

Communication Technology

transfer and disseminate
digital information

INFORMATION COMMUNICATION TECHNOLOGY (ICT)

Information knowledge

Technology use of computers and communication

“the integration of computing, networking, and information processing technologies and their applications.”

ICT includes...

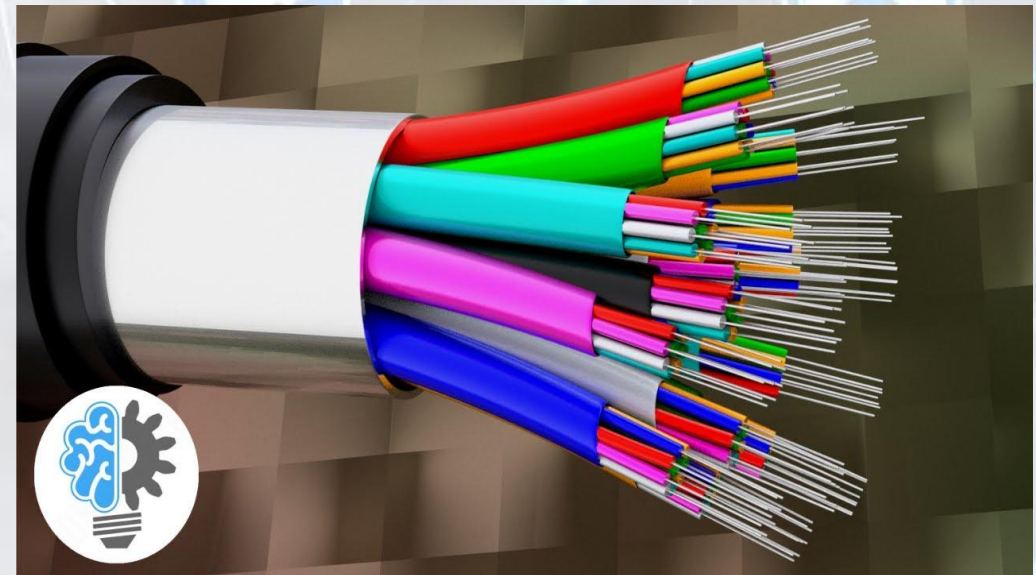
- computer hardware, software, application of telecommunication technologies
- projection devices,
- Local Area Networks (LAN)
- Wide Area Networks (WAN)
- digital cameras



ICT includes...

- Compact Disks (CDs)
- Digital Video Disks (DVDs)
- cell phones
- satellites
- fiber optic

Fiber Optic Cable



Other definitions of ICT

According to the **Encyclopedia of Computer Science**, “Information Communication Technology (ICT) is an imprecise term frequently applied to broad areas of activities and technologies associated with the use of computers and communications”.

Other definitions of ICT

American Library Association (1983) defined information communication technology (ICT) as the application of computers and other technologies to the acquisition, organization, storage, retrieval, and dissemination of information. The computers are used to process and store data, while telecommunication technology provides information communication tools, which make it possible for users to access databases and link them with other computer networks at different locations.

Characteristics of ICT

Most interactive

Fewer errors

Effectiveness

Customized

Personalized



Characteristics of ICT

Faster

Less paperwork

Cheaper

Efficiency

Fewer steps

Lower costs



Characteristics of ICT

New product

Innovation

New
Technologies

History of ICT in the Philippines



Before ICT in the PH

Before 1928, telecommunications in the Philippines was segmented. You could only call people within your own small city.



1928: Telecoms in the Philippines

1928 American-owned PLDT was incorporated and given franchise to establish and operate telephone services in Philippines. Small phone companies in the provinces were to speed up the rollout process.

History of ICT in the Philippines



the PH

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1928: Telecoms in the Philippines

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1968 Filipinos Connecting World

PLDT became a Filipino-controlled corporation with the acquisition of Cojuangco.

History of ICT in the Philippines



PLDT in the Philippines

PLDT was incorporated and given the right to plan, build, and operate telephone services in the Philippines. The companies in the provinces were acquired in a process.



1968 Filipinos Connecting Filipinos to the World

PLDT became a Filipino-controlled corporation bought by Ramon Cojuangco.



1987: Cellular Telephone

PLDT establishes the country's first cellular telephone network.

History of ICT in the Philippines



Connecting Filipinos to the

controlled corporation bought by Ramon



1987: Cellular Telephone Network

PLDT establishes the country's first cellular telephone network.



1993: Birth of the Philippine

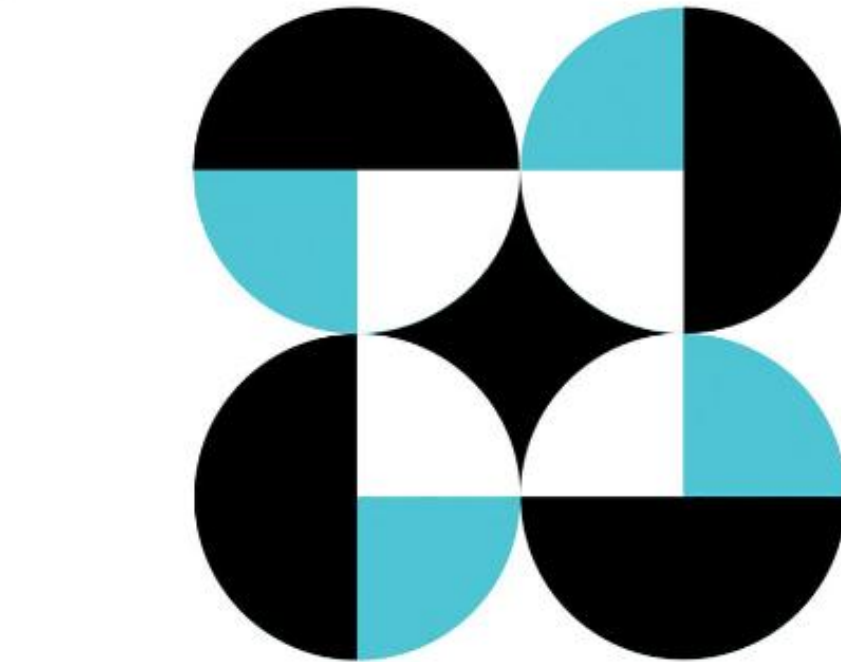
With the support of the Department of Science and Technology and the Industrial Research Foundation, the PHNET project (now PHNET) was born.

History of ICT in the Philippines



Phone Network

first cellular telephone network.



1993: Birth of the Philippine Internet

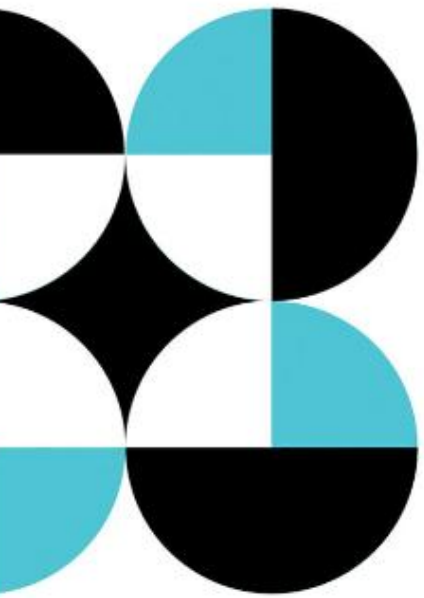
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1994: Our First Internet Connection

Benjie Tan, who was working for ComNet, made the **first connection to the Internet** at a PLDT office in Manila City.

History of ICT in the Philippines



the Philippine Internet

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1994: Our First Internet Connection

Benjie Tan, who was working for ComNet established **Philippines' first connection to the Internet** at a PLDT network center in Makati City.



2013: The Massive Growth of the Country's Mobile Network

Our Mobile Cellular Subscriptions reach

History of ICT in the Philippines



Internet Connection

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2013: The Massive Growth of the Country's Mobile Networks

Our Mobile Cellular Subscriptions reach 102 million.

2014: Fast

The Philippines m
last five years wi

Number of Philip
of 100 million.

History of ICT in the Philippines

2014: Fastest Growing Connections

The Philippines named fastest growing internet population in the last five years with a growth of 531%

Number of Philippine Internet users at 38 million out of a population of 100 million.

History of ICT in the Philippines

2017: Internet Connection Population Penetration

119M Mobile Phone Subscriptions in the Country (117% Penetration Rate)

95% Prepaid

Greater 10% Broadband Subscription

55% Mobile Broadband Subscription

80% Subscribed to the Lowest Speed Plans (1–3 MBPS)

TODAY IN ICT

The time we spent online



3.2 hours on mobile phones



TODAY IN ICT

The time we spent online



5.2 hours on desktops/
tablets

TODAY IN ICT

TOP FILIPINO ONLINE ACTIVITIES



FACEBOOK 47%



ONLINE VIDEOS /
YOUTUBE 19%



GAMING 15%



SEARCH 13%



SHOPPING 29%

CURRENT STATE OF ICT

Convergent Technologies

This is when multiple different devices combine into a single gadget because of technological advancement.



Drivers of Technology Convergence

Michael Stankosky, DSc

- ✓ Information technology and digitization
- ✓ Globalization
- ✓ Sophistication of services
- ✓ Pressures for increased functionality
- ✓ drive to find solutions to the complexity to our industrial system
- ✓ Increased process orientation
- ✓ Sustainability

Examples of Technology Convergence

Artificial Intelligence (AI)

Smartphones/Smart Tv

Laptops/Computers

Online banking apps

Social Media



Pioneers and Leaders of Converging Technology

- Apple
- Adobe
- Netflix
- Amazon Prime
- Microsoft
- Google



Benefits of Convergent Technology

- Time-saving and cost-saving devices.
- Improving human performance
- Allowing and encouraging new ways to communicate.
- Bigger effectiveness of a single piece of technology.

Benefits of Convergent Technology

- One set of infrastructure which is cheaper to operate.
- Watching media on a wide range of devices.
- Broader availability.

What is the Future of Converging Technology?

- Quantum Computing
- Truly integrated health research/care
- Whatever you can imagine

CURRENT STATE OF ICT

Social Media

These are software, applications or platforms that allow users to communicate in an online social communities or network.



CURRENT STATE OF ICT

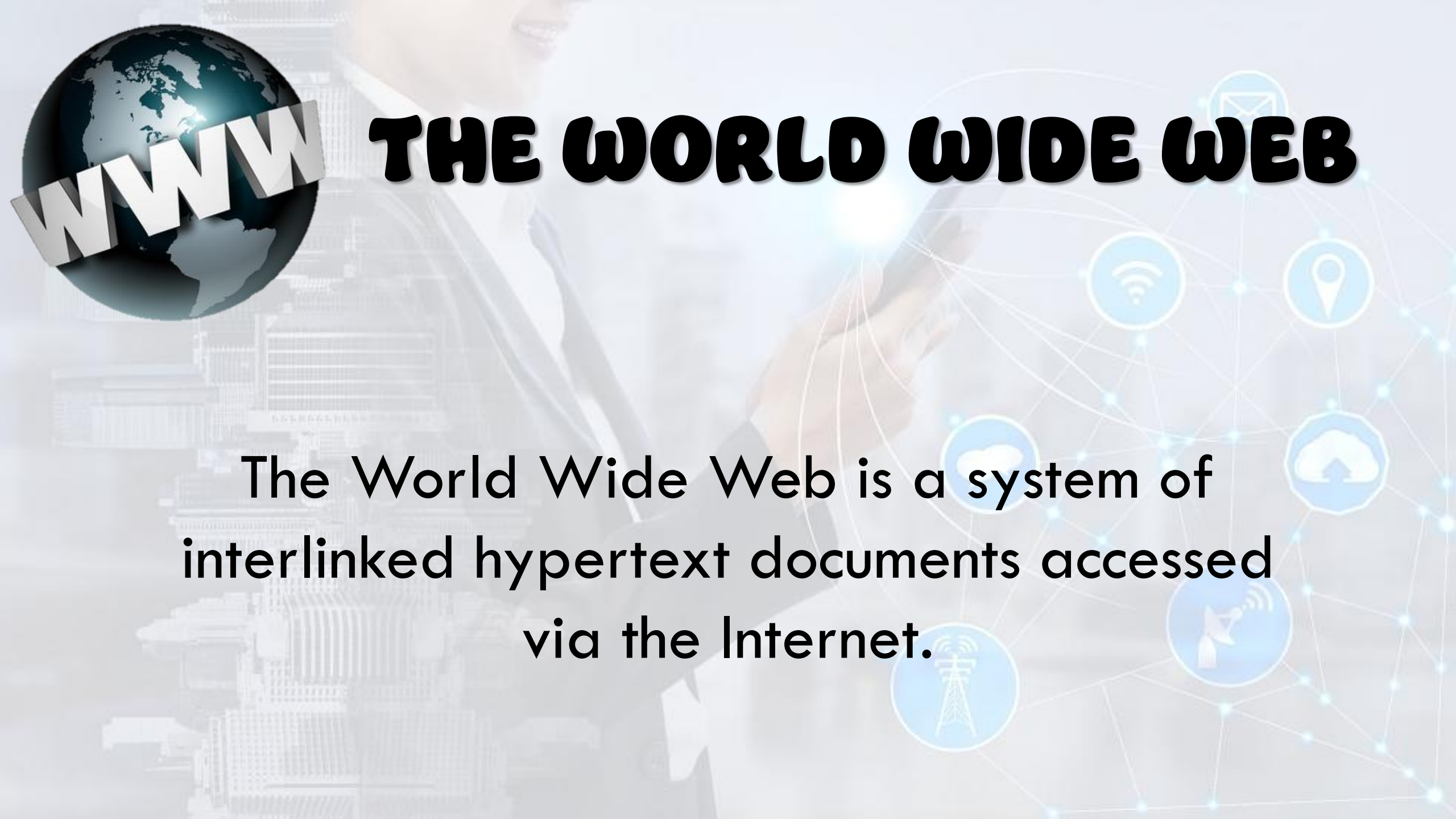
Mobile Technologies

- These are devices that are easily carried and allows fast means of communication.
- These devices from the past were originally bulk, but due to the technological advancement, they become more smaller and more compact.

Evolution of Computer







THE WORLD WIDE WEB

The World Wide Web is a system of interlinked hypertext documents accessed via the Internet.

Is Internet and World Wide Web the same?



THE WORLD WIDE WEB

system that enables people
to access information over
the internet

software aspect: it relates
to the Protocols or HTTP
(HyperText Transfer
Protocol)

INTERNET

actual interconnection
of computers and
other networks

it relates to computer
networks, connections and
infrastructures

THE BIRTH OF WORLD WIDE WEB



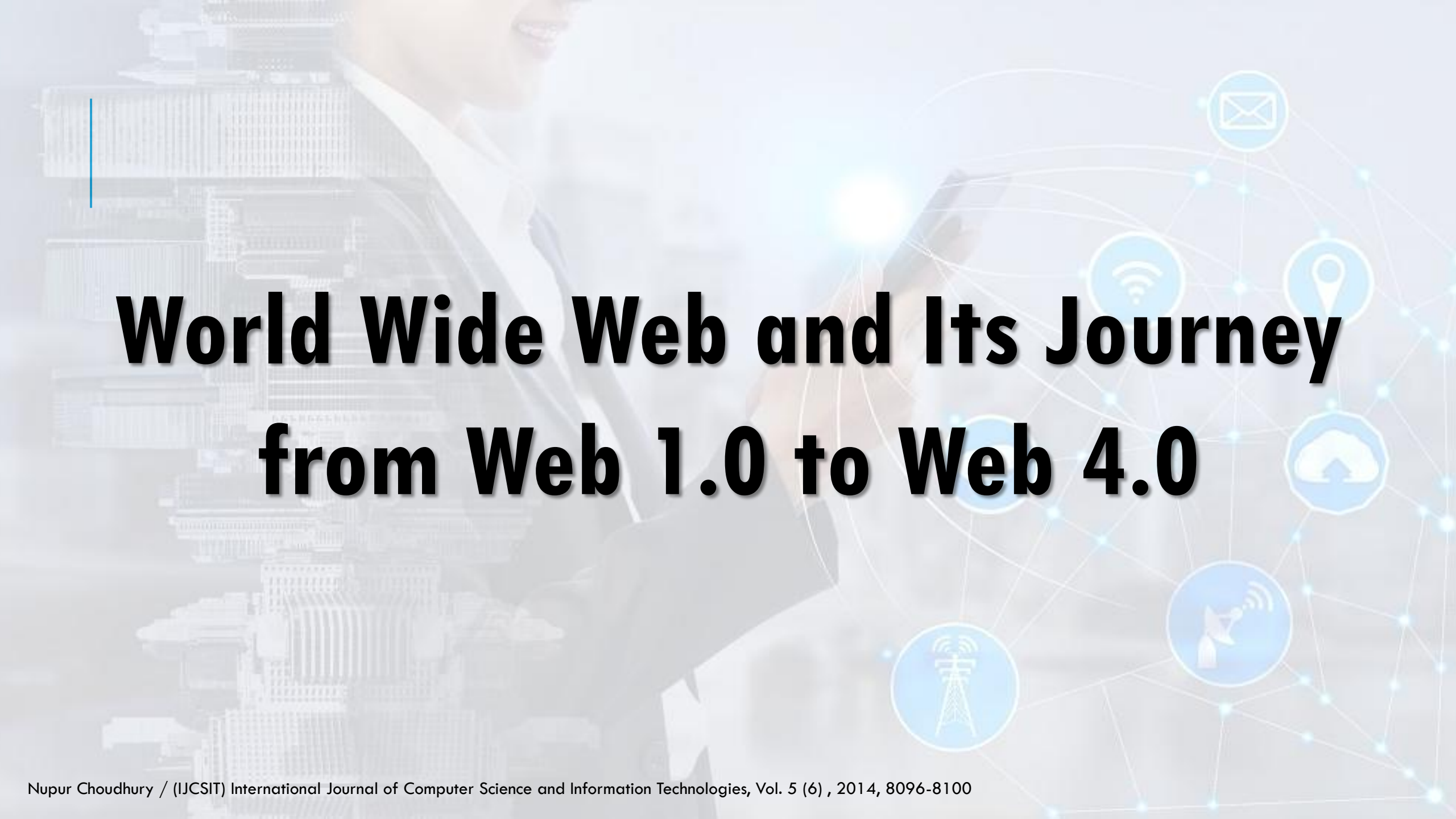
Tim Berners-Lee, a British scientist, invented the World Wide Web (WWW) last March 12, 1989, while working at CERN.

THE BIRTH OF WORLD WIDE WEB

- The web was originally conceived and developed to meet the demand for automated information-sharing between scientists in universities and institutes around the world.
- Berners- Lee and Belgian computer scientist Robert Cailliau proposed in 1990 to use hypertext “to link and access information of various kinds as a web of nodes in which the user can browse at will”

HYPERTEXT TRANSFER PROTOCOL

- are the system or procedures that enables the transfer of information in the internet web services, applications and platforms
- used to load webpages using hypertext links

The background features a person in a dark suit and white shirt, holding a smartphone. A bright light emanates from the phone's screen. Overlaid on the right side is a network diagram with blue circular nodes containing icons: an envelope, a Wi-Fi symbol, a location pin, a cloud with an upward arrow, a satellite dish, and a radio tower. These nodes are connected by thin white lines. The overall theme is digital connectivity and technology.

World Wide Web and Its Journey from Web 1.0 to Web 4.0

WEB 1.0

- first implementation of the web and it lasted from 1989 to 2005
- web of information connections
- According to the innovator of World Wide Web, Tim Berners-Lee considers the Web as **“read-only”** Web

WEB 1.0

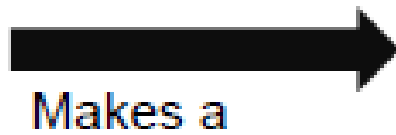
- It provides very little interaction where consumer can exchange the information together, but it was not possible to interact with the website.
- First generation Web was era static pages and content delivery purpose only.
- early web allowed us to search for information and read it

WEB 1.0: *Limitations*

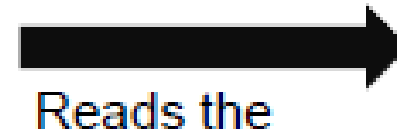
- The Web 1.0 pages can only be understood by humans (web readers) they do not have machine compatible content.
- The web master is solely responsible for updating users and managing the content of website.
- Lack of Dynamic representation i.e., to acquire only static information, no web console were available to performing dynamic events.

WEB 1.0

Creator



WEB 1.0
CONTENT



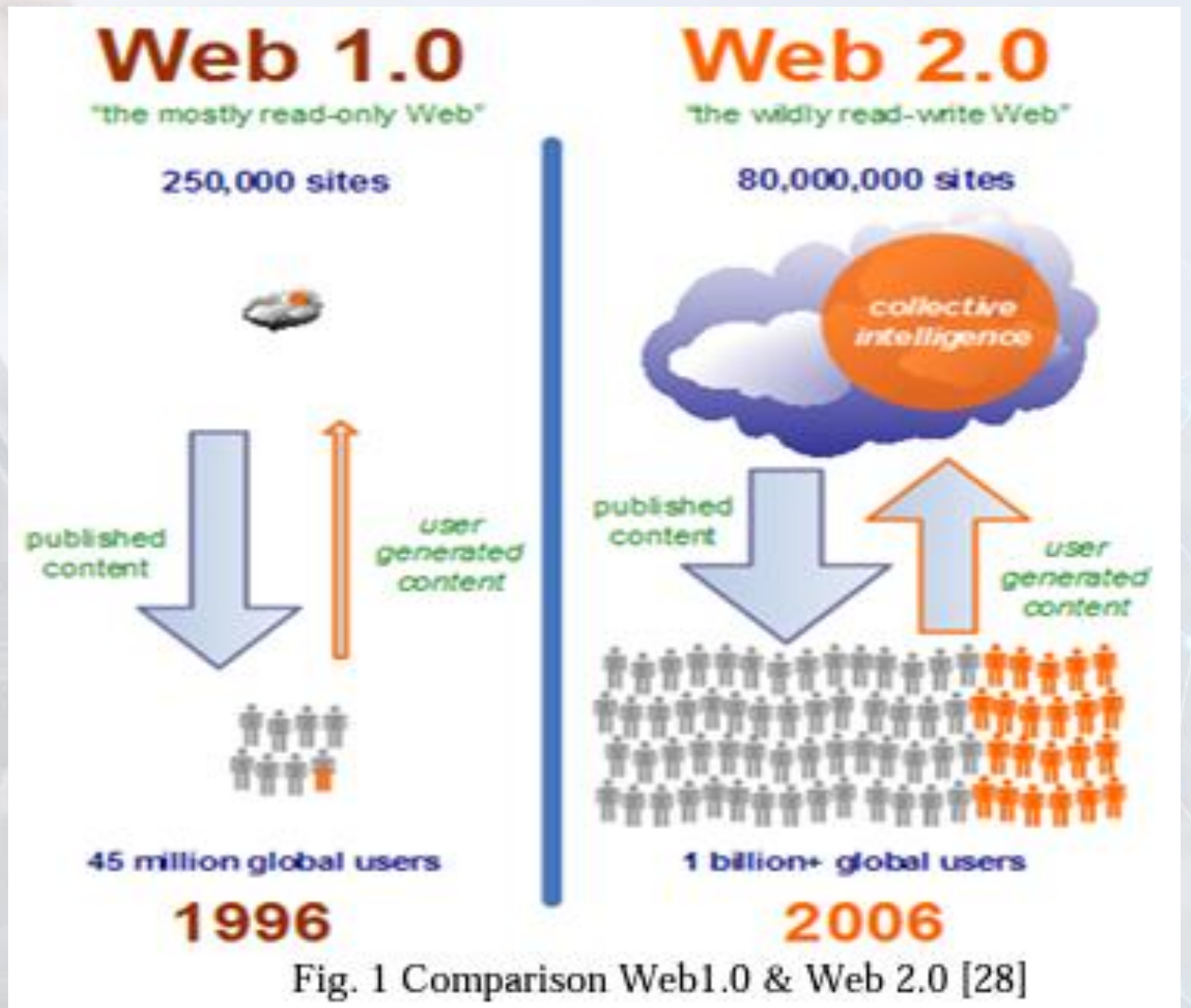
User

According to the inventor of the world wide web, Tim Berner Lee, web 1.0 is “Read only web” which means the user can only read whatever online content the creator has made.

WEB 2.0

- second generation of web
- coined by Tim O'Reilly
- defined by Dale Dougherty in 2004 as a read-write web.
- facilitates major properties like participatory, collaborative, and distributed practices which enable formal and in-formal spheres of daily activities on going on web

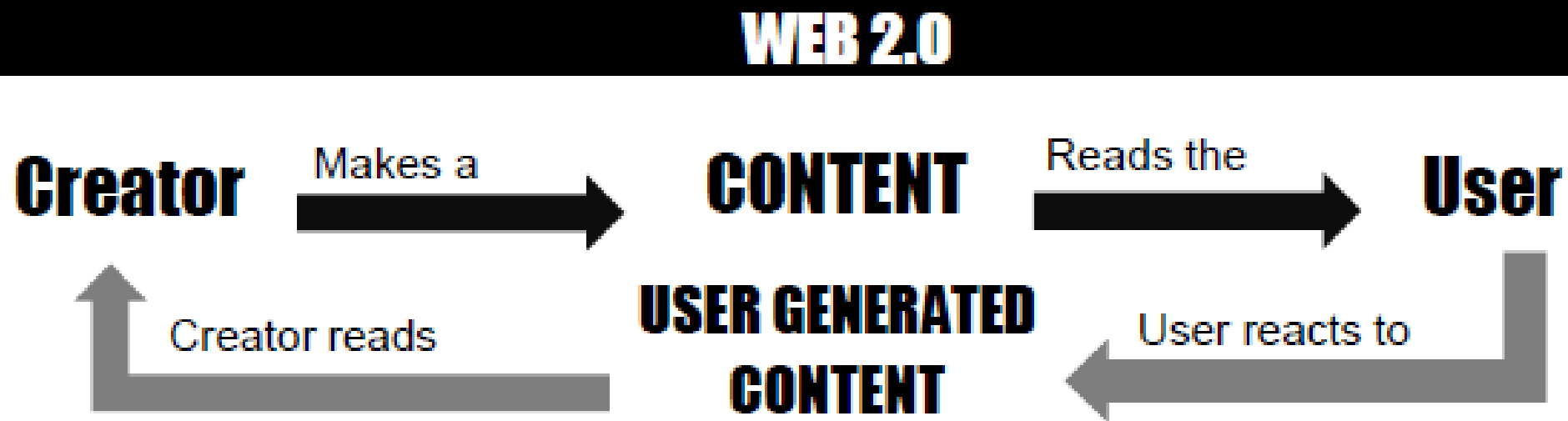
WEB 2.0



WEB 2.0: *Limitations*

- Constant iteration cycle of Change and Updates to services
- Ethical issues concerning build and usage of Web 2.0
- Interconnectivity and knowledge sharing between platforms across community boundaries are still limited

WEB 2.0



Web 2.0 is “Read-Write”, which refers for the ability of the user to add content thru posting, sharing, commenting or rating. Web 2.0 allows interaction not only in between the user and the creator but also to other users. Examples are sites like Facebook, YouTube, Twitter and Reddit.

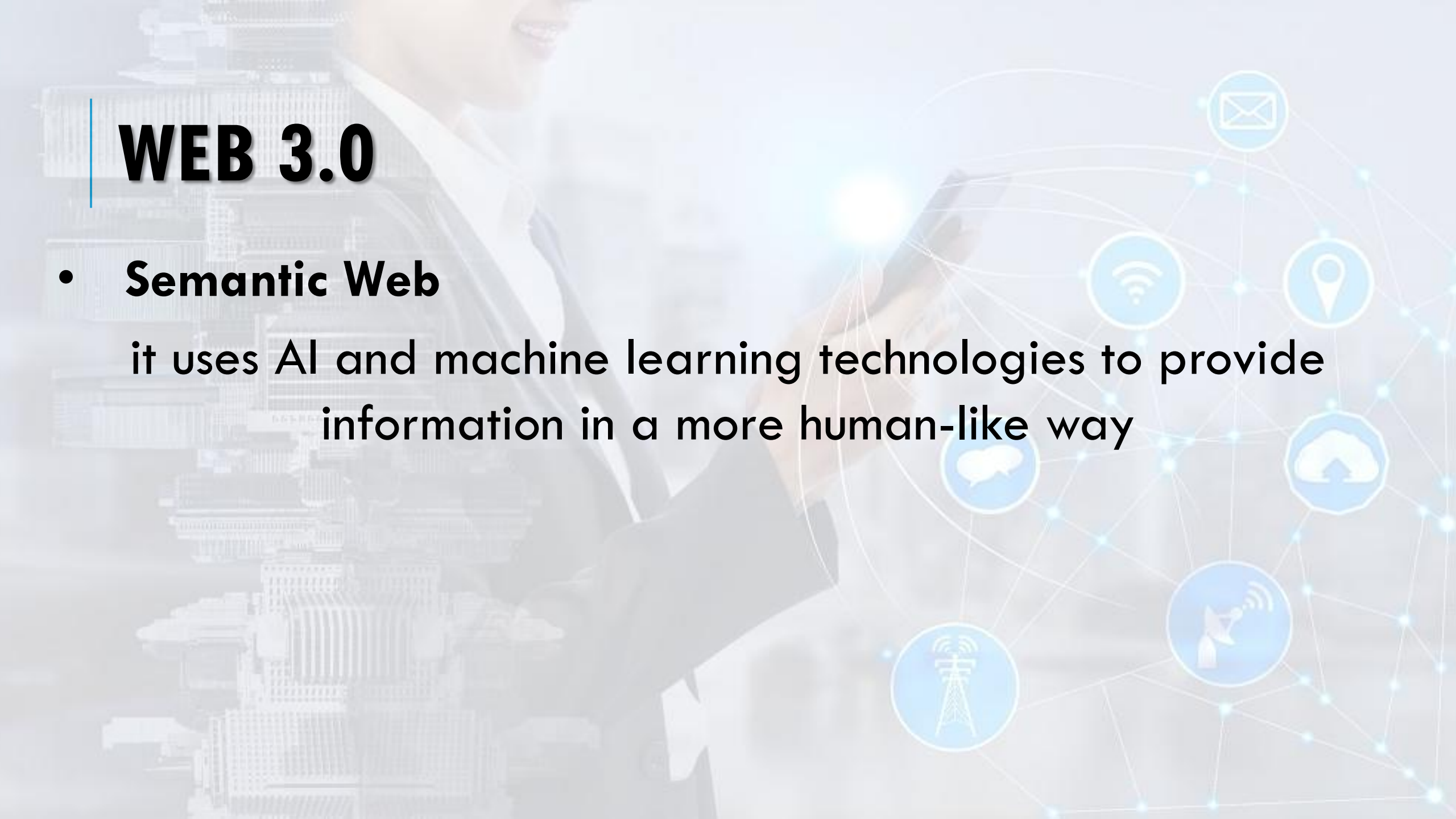
WEB 3.0

- first coined by John Markoff of the New York Times and he suggested web 3.0 as third generation of the web in 2006
- “executable web”
- capability of accepting a request from the user, will execute a solution or suggestion considering the behavioral data the user

WEB 3.0

- **Semantic Web**

it uses AI and machine learning technologies to provide information in a more human-like way



WEB 3.0: *Challenges*

- **Vastness**

The World Wide Web contains many billions of pages. Redundancy in Data may occur which has not yet been able to eliminate all semantically duplicated terms.

WEB 3.0: *Challenges*

- **Vagueness**

This arises from the vagueness of user queries, of concepts represented by content providers, of matching query terms to provider terms and of trying to combine different knowledge bases with overlapping but subtly different concepts.

WEB 3.0: *Challenges*

- **Inconsistency**

These are logical contradictions which will inevitably arise during the development of large ontologies, and when ontologies from separate sources are combined.

WEB 3.0: *Challenges*

- **Deceit**

This is when the producer of the information is intentionally misleading the consumer of the information.

WEB 3.0

WEB 3.0

User

Request suggestion
for a restaurant



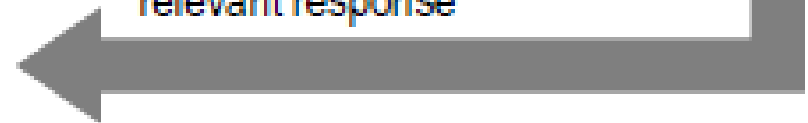
**WEB 3.0
ENABLED
DEVICE**

Scans the internet
based on various data

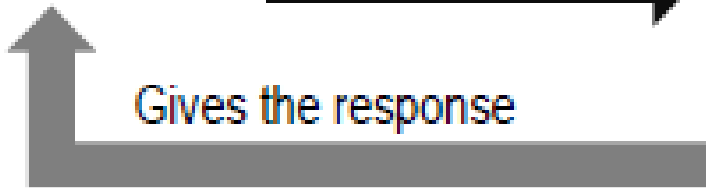


Internet

Filters the internet for the most
relevant response



Gives the response



WEB 4.0 and Future Web

- can be considered as an Ultra-Intelligent Electronic Agent, symbiotic web and Ubiquitous Web
- Powerful as human brain, progress in the development of telecommunications, advancement on nanotechnology in the world and controlled interfaces using web 4.0.

WEB 4.0 and Future Web

- Web 4.0 will be read write concurrency web
- It ensures global transparency, governance, distribution, participation, collaboration into key communities such as industry, political, social and other communities.

WEB 4.0 and Future Web

- WebOS will be such as a middleware in which will start functioning like an operating system.
- WebOS will be parallel to the human brain and implies a massive web of highly intelligent interactions.

Can ICT replace humans?



Activity 1

Using the table on the next slide, think of things you're doing/you've done that involves ICT technologies. Then beside it, research or ask older people how were their lives then without ICT technologies. Then create a generalization or conclusion based on the information you have collected.

Activity 1

What is it now with ICT?	What was it then without ICT?
Generalization/Conclusion:	